

Supplementary appendix

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Supplement to: Siderowf A, Concha-Marambio L, Lafontant D-E, et al. Assessment of heterogeneity among participants in the Parkinson's Progression Markers Initiative cohort using α -synuclein seed amplification: a cross-sectional study. *Lancet Neurol* 2023; **22**: 407–17.

Table S1: Biomarker characteristics by α Syn-SAA status for Sporadic PD, LRRK2 PD, GBA PD, and HC

Variable	Sporadic PD		LRRK2 PD		GBA PD		HC	
	SAA+ (N = 348)	SAA- (N = 25)	SAA+ (N = 83)	SAA- (N = 40)	SAA+ (N = 47)	SAA- (N = 2)	SAA+ (N = 6)	SAA- (N = 157)
Mean striatum SBR	1.35 [1.14-1.62]	1.26 [1.04-1.84]	1.24 [0.97-1.54]	1.39 [1.06-1.49]	1.16 [0.92-1.56]	0.80 [N/A]	2.45 [2.09-3.16]	2.44 [2.19-2.90]
CSF abeta (< 683)	104 (32%)	8 (35%)	27 (33%)	13 (34%)	23 (50%)	0 (N/A)	0 (N/A)	41 (27%)
CSF tau (< 266)	303 (93%)	22 (92%)	77 (93%)	32 (82%)	44 (96%)	2 (N/A)	5 (N/A)	135 (87%)
CSF α syn (< 1000)	71 (22%)	3 (13%)	11 (24%)	2 (10%)	4 (50%)	N/A	1 (N/A)	24 (15%)
Serum NfL	11.50 [8.30-15.80]	13.85 [11.60-19.10]	10.70 [8.77-14.95]	18.60 [10.50-30.00]	12.90 [9.58-17.50]	26.70 [N/A]	12.45 [9.14-16.10]	10.30 [7.09-14.80]
	<i>Difference = -0.19 (-0.40, 0.01)*</i>							
Total di-18:1 BMP	3.7 [2.1-6.8]	2.9 [1.8-5.4]	10.9 [7.7-20.7]	15.0 [6.5-29.1]	5.9 [2.4-9.5]	12.9 [N/A]	5.6 [2.7-11.7]	3.5 [1.9-6.4]

Summary statistics: Median [Q1-Q3]; N (%); Median Differences (95% CI) centered below relevant comparisons.

Median differences with 95% confidence intervals were used for within group comparisons of α Syn-SAA status for Mean striatum SBR, Serum NfL, and Total di-18:1 BMP.

* The 95% confidence interval for the mean difference in log serum NfL between α Syn-SAA negative and α Syn-SAA positive participants using an age- and sex-adjusted linear regression model.

Odds ratio with 95% confidence intervals were used for within group comparisons of α Syn-SAA status.

Comparisons for GBA and HC were not performed due to low sample size.

Table S2: Clinical Characteristics for NMC, Hyposmia, and RBD

<i>Variable</i>	<i>LRRK2</i>		<i>GBA</i>		<i>Hyposmia</i>		<i>RBD</i>	
	<i>SAA+ (N = 14)</i>	<i>SAA- (N = 145)</i>	<i>SAA+ (N = 11)</i>	<i>SAA- (N = 140)</i>	<i>SAA+ (N = 16)</i>	<i>SAA- (N = 2)</i>	<i>SAA+ (N = 28)</i>	<i>SAA- (N = 5)</i>
Age (yrs)	63.3 [57.7-76.4]	61.5 [56.0-67.1]	61.1 [60.5-70.8]	62.1 [57.3-66.5]	66.4 [63.4-70.1]	78.2 [N/A]	69.7 [66.3-73.1]	70.8 [62.4-77.0]
Male sex	7 (50%)	59 (41%)	5 (45%)	63 (45%)	10 (63%)	2 (N/A)	23 (82%)	5 (N/A)
Race (% White)	12 (92%)	141 (99%)	11 (100%)	135 (97%)	15 (94%)	2 (N/A)	27 (100%)	4 (N/A)
MDS-UPDRS I	3 [2-6]	4 [2-6]	3 [1-9]	4 [2-7]	4 [2-7]	6 [N/A]	9 [4-11]	6 [4-7]
MDS-UPDRS II	1 [0-2]	0 [0-1]	0 [0-2]	0 [0-1]	2 [0-3]	2 [N/A]	2 [0-5]	1 [1-3]
MDS-UPDRS III	2 [1-4]	2 [0-4]	0 [0-4]	1 [0-4]	1 [0-5]	5 [N/A]	3 [2-6]	5 [3-8]
Total MDS-UPDRS	7 [4-12]	6 [4-12]	3 [2-24]	7 [3-13]	8 [4-17]	13 [N/A]	14 [8-19]	14 [8-18]
% Hyposmic (<=15%)	6 (43%)	15 (10%)	6 (55%)	13 (9%)	16 (100%)	1 (N/A)	26 (96%)	1 (N/A)
DAT Lowest Putamen SBR	0.83 [0.64-1.06]	0.94 [0.82-1.10]	1.05 [0.92-1.24]	1.13 [0.99-1.28]	0.62 [0.52-0.72]	0.72 [N/A]	0.52 [0.44-0.64]	0.61 [0.45-0.68]

Summary statistics: Median [Q1-Q3]; N (%)

Statistical analyses were not provided due to small numbers in some groups.

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